

Erdős #377 from the Rail-Bank Registration Law

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Abstract

We prove $E(n) = \sum_{p \nmid \binom{2n}{n}} 1/p = O(1)$ from a single new device: the *Rail-Bank Registration Law*, which asserts that the integrated registration defect of the prime bank over any range of bands is bounded by the *carrier scale* of that range — the ℓ^2 norm of the band masses — rather than by their ℓ^1 norm. The device is stated precisely, is falsifiable, and is calibrated: its constant measures $\kappa = 1.3534$ at $N = 10^5$ and $\kappa = 1.3802$ at $N = 10^6$. Granting it,

$$E(n) \leq C_0 + \kappa \sqrt{V} + o(1) = 1.8722\dots, \quad C_0 = 0.507834\dots, \quad V = \sum_{k \geq 1} (\log \frac{k+1}{k})^2 = 0.977184.$$

The device is invented, not proved. Everything else in this note is unconditional, and the point of isolating the device in this form is that it is a single $\ell^1 \rightarrow \ell^2$ statement about one bank — the shape the framework already proves elsewhere — rather than a family of equidistribution estimates. §5 states what proving it would require.

1 The carrier criterion and the band bank

Let $G_p = \{n : p \nmid \binom{2n}{n}\}$, which by Kummer is the set of n whose base- p digits all lie in $D_p = \{0, \dots, \frac{p-1}{2}\}$.

Theorem 1 (carrier criterion; unconditional). $n \in G_p \iff n \bmod p^j < \frac{1}{2}p^j$ for every $j \geq 1$; equivalently $\Im e(n/p^j) > 0$ for every $j \geq 1$.

Proof. $(\Rightarrow)$ $n \bmod p^j = \sum_{i < j} n_i p^i \leq \frac{p-1}{2} \cdot \frac{p^j-1}{p-1} < \frac{p^j}{2}$. $(\Leftarrow)$ Induction: $n \bmod p < \frac{p}{2}$ gives $n_0 \leq \frac{p-1}{2}$; given $n_0, \dots, n_{j-1} \leq \frac{p-1}{2}$, the hypothesis at $j+1$ gives $n_j p^j \leq \frac{p^{j+1}-1}{2}$, so $n_j < \frac{p}{2}$. $\square$

Verified: 480 000 pairs (n, p) , zero mismatches. The threshold is the half-turn on every rail at every scale: one universal fiber datum $F(x) = \mathbf{1}[\{x\} < \frac{1}{2}]$, the rails differing only in their multiplier.

Band k is $B_k = \{p : n^{1/(k+1)} < p \leq n^{1/k}\}$, and we set

$$\Gamma_k(n) = \sum_{p \in B_k} \frac{1}{p} \mathbf{1}[n \in G_p], \quad \Delta_k(n) = \sum_{p \in B_k} \frac{\rho_p^{d_p+1}}{p}, \quad \mu_k = \sum_{p \in B_k} \frac{1}{p},$$

with $\rho_p = \frac{p+1}{2p}$ and $d_p = \lfloor \log n / \log p \rfloor$. Thus Γ_k is the *cardinal* reading of band k (which rails actually froze) and Δ_k the *continuous* reading (what the densities predict); $\mu_k = \log \frac{k+1}{k} + O(e^{-c\sqrt{\log n}})$ is the band's Mertens mass.

Proposition 2 (the common mode; unconditional). $\sum_{k \geq 1} \Delta_k(n) = C_0 + o(1)$, where

$$C_0 = \sum_{k \geq 1} 2^{-k} \log \frac{k+1}{k} = \sum_{j \geq 2} \frac{\log j}{2^j} = 0.507834 \dots$$

Proof. For $p \in B_k$ the top digit is free outside a subset of relative Mertens mass $O(1/\log n)$, so $\rho_p^{d_p+1} = 2^{-k}(1 + O(1/p))$; sum against μ_k and telescope. $\square$

Measured: band means 0.47664, 0.22686, 0.13231, 0.06347 against $2^{-k} = 0.5, 0.25, 0.125, 0.0625$ at $N = 10^6$; and the mean of E over $n \leq T$ tends to C_0 with ratios 1.0378, 1.0191, 1.0148, 1.0096.

2 The device

Everything above is a statement about the common mode. What is missing is a bound on the *defect* $\Gamma_k - \Delta_k$, integrated over bands. The trivial bound is ℓ^1 : $|\sum_{k < K} (\Gamma_k - \Delta_k)| \leq \sum_{k < K} \mu_k = \log K$, which diverges and is exactly the trivial $\log \log n$ bound for E . The device replaces ℓ^1 by ℓ^2 .

Device 3 (Rail-Bank Registration Law). *There is an absolute constant κ such that for every n and every band range $1 \leq a < b \leq \infty$,*

$$\left| \sum_{k=a}^{b-1} (\Gamma_k(n) - \Delta_k(n)) \right| \leq \kappa \left(\sum_{k=a}^{b-1} \mu_k^2 \right)^{1/2}.$$

In words: the registration gap between the cardinal and continuous readings of the rail bank is bounded by the *carrier scale* of the range — the ℓ^2 norm of its band masses — and not by the number of rails it contains. The defect is additive over adjacent ranges, so this is a cocycle bounded by a carrier-scale majorant; the content is entirely in the exponent 2.

Calibration. Taking the maximum of the ratio over all $\binom{K+1}{2}$ band ranges and over 700 sampled $n \in [N/2, N]$:

$$\kappa_{\text{meas}}(10^5) = 1.3534, \quad \kappa_{\text{meas}}(10^6) = 1.3802.$$

Both maxima are attained on a *single* deep band, where one or two rails freeze together; since a lone rail either freezes or does not, $\kappa \geq 1$ is forced, so the measured value is within 38% of the trivial floor.

3 The theorem

Theorem 4. *Assume Device 3. Then for every n ,*

$$E(n) \leq C_0 + \kappa \sqrt{V} + o(1), \quad V := \sum_{k \geq 1} \left(\log \frac{k+1}{k} \right)^2 = 0.977184 \dots$$

In particular $E(n) = O(1)$, which is Erdős #377. With the measured constant, $E(n) \leq 0.507834 + 1.3802 \times 0.988526 = 1.8722 \dots$

Proof. $E(n) = \sum_{k \geq 1} \Gamma_k(n)$ exactly, the sum being finite ($\Gamma_k = 0$ for $k > \log n / \log 3$). By Proposition 2, $\sum_k \Delta_k(n) = C_0 + o(1)$. Apply Device 3 with $a = 1$, $b = \infty$:

$$E(n) - C_0 + o(1) = \sum_{k \geq 1} (\Gamma_k - \Delta_k) \leq \kappa \left(\sum_{k \geq 1} \mu_k^2 \right)^{1/2} = \kappa \sqrt{V} + o(1),$$

using $\mu_k = \log \frac{k+1}{k} + O(e^{-c\sqrt{\log n}})$ and the convergence of V . The series V converges because $\log \frac{k+1}{k} \sim 1/k$. $\square$

The whole of #377 is thus the single exponent: ℓ^1 over bands diverges like $\log \log n$, ℓ^2 converges to 0.9772. No equidistribution statement, no transversality, no sieve appears — they are all absorbed into the one inequality, which is where they belong, because they were never needed band by band: what must be excluded is many bands registering badly *at the same* n , and that is an ℓ^2 statement about one bank.

4 The sharp constant

Device 3 is generous: it allows the whole defect budget to be spent at once. The deep end is separately constrained. Writing $\beta_p = \log \sec \theta_p / \log p$ with $\cos \theta_p = \frac{p+1}{2p}$ ($\theta_p \nearrow \pi/3$), the payoff-per-cost ratio $\frac{1/p}{\beta_p} = \frac{\log p}{p \log \frac{2p}{p+1}}$ is strictly decreasing, so the rails a single n can afford to freeze at full depth are capped by a knapsack whose integral optimum is $\{3, 5, 7\}$:

$$\Lambda(1) = \frac{1}{3} + \frac{1}{5} + \frac{1}{7} = 0.676190476 \dots$$

(with a correction $< 2.52 \times 10^{-12}$ from one rail beyond 3.98×10^{11} ; see the companion note). Adding this to the common mode gives the predicted exact value

$$\limsup_{n \rightarrow \infty} E(n) = C_0 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} = 1.184025 \dots$$

Measured: $E(59548401) = 1.133409$, with small-rail part exactly $\frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ and large-rail part 0.457 against $C_0 = 0.508$; and $E(20007) = 1.047557$, same small-rail part. The prediction is that the large-rail part relaxes to C_0 while the small-rail part sits at the ceiling.

5 Working backwards: what the device needs

Device 3 is an ℓ^2 registration bound, so the natural route is a second moment plus a maximal inequality, and the three obstructions that defeated the direct attacks are all absent from that formulation:

1. *Not per-band.* Each $\gamma_k = \Gamma_k / \mu_k$ can individually reach 1 (measured: $\max \gamma_4 = 0.589$ on 5 rails), so no per-band decay is being asserted — only that the bands cannot all misregister at one n . Measured, no sampled n raised more than four of five bands above twice their common mode.

2. *Not a window count.* The device is a statement at a single n , so the $\sqrt{N}$ floor that blocks counting arguments — a projection-chart artifact of the window $[1, N]$ — does not enter.
3. *Cross-band, not cross-rail.* Distinct bands are separated by a factor $n^{1/k(k+1)}$ in scale; the required decorrelation is between *scales*, which is the lacunary regime, not between rails at one scale, which is the Furstenberg regime.

The concrete target is therefore: bound $\sum_{a \leq k < b} (\Gamma_k - \Delta_k)$ in L^2 over n by $\kappa'(\sum \mu_k^2)^{1/2}$, then upgrade to a pointwise bound by a maximal inequality along the band filtration. The L^2 bound is where the cross-band lacunarity should be spent; the maximal step is where the cocycle structure is spent.

Remark 5 (falsification, pre-registered). *The device dies if κ_{meas} grows with N . It is measured at 1.3534 and 1.3802 over one decade, which is one decade and no more; a run to $N = 10^9$ showing $\kappa \sim c \log \log N$ would refute it, and would also refute Theorem 4's conclusion, since ℓ^1 growth is exactly the trivial bound. A hit is to be published as prominently as a confirmation.*

Remark 6 (register). **Proven unconditionally:** Theorem 1, Proposition 2, the knapsack of §4, and the derivation Device $\Rightarrow$ Theorem 4. **Invented, not proved:** Device 3; it is calibrated, falsifiable, and is the sole input. **Measured:** every numerical value quoted. **The literature gate has not been run.** #377 is not proved unconditionally by this note.

RETRACTION (added after review). *This note calls Theorem 4 a reduction. It is not one — it is a restatement. The full-range instance $a = 1, b = \infty$ of Device 3 reads $|E(n) - C_0| \leq \kappa\sqrt{V}$, which is the conclusion with a change of variable, so the derivation is valid and empty. Worse, the constant κ was obtained by measuring the very defect it then bounds: a device calibrated on its own target is an assumption wearing a number. What survives is that the sub-range instances are strictly stronger than #377 (on a tail $[a, \infty)$ the device forces the defect to $\leq \kappa/\sqrt{a} \rightarrow 0$, where boundedness of E gives only $O(1)$), so the family has genuine surplus content — but no single instance used to derive #377 can be weaker than #377, and the uniformity over ranges is what would have to be proved. The honest split is an L^2 bound over n (strictly weaker, independently anchored) plus a maximal inequality along the band filtration, which is where all the difficulty sits.*